Wednesday, April 28, 1999 (LSC 4258)

12:15—1:45 **Registration**

1:45 **Introduction to the In-House Convention** —Richard Brown

2:15—3:00 **Session 1.** Chair: Fatina Elkurdi
T1) Dean Tripp
T2) Ivan Kiss
T3) Jana Podhorna

3:00—3:30 **Break**

3:30—4:30 **Session 2.** Chair: Melanie McFadyen
T4) Lachlan McWilliams
T5) John Barresi
T6) Lianne Stanford
T7) William Brown
T8) Shelley Adamo

Thursday, April 29, 1999 (LSC 4258)

9:30—10:30 **Session 3.** Chair: Larissa Hausmanis
T9) Cathy Forestell
T10) Kelly Shoemaker
T11) Sonia Greaven
T12) Nadine Rossy

10:30—11:00 **Coffee Break**

11:00—12:15 **Session 4.** Chair: Rochelle Currie
T13) Karen Jaskie
T14) Mary-Ellen Large
T15) Deanna Braaksma
T16) Nicole Blanchet
T17) Jason Ivanoff

12:15—1:45 **Lunch Break and Posters 1-8 (Lounge)**

1:45—3:00 **Session 5.** Chair: Patti Guidry
T18) Mary Ann Campbell
T19) Shana Nichols
T20) Sharon Clark
T21) Randy Lynn Newman
T22) Fernando Larrea

3:00—3:30 **Coffee Break**

3:30—5:00 **Keynote Speech:** Dr. Peter Jusczyk

Friday, April 30, 1999 (LSC 4258)

9:30—10:30 **Session 6.** Chair: Jennifer Stapleton
- T23) Ryan D'Arcy
- T24) Ray Klein
- T25) Jeff Hancock
- T26) Susan Boehnke

10:30—11:00 **Coffee Break**

11:00—12:15 **Session 7.** Chair: Jennifer Stapleton
- T27) Darcy Santor
- T28) Shui-Wang Ying
- T29) Alfredo Villarroel
- T30) Leticia Materi
- T31) Steve Shaw

12:15—1:45 **Lunch Break and Posters 9-14 (Lounge)**

1:45—3:00 **Session 8.** Chair: Sarah Giles
- T32) Bev Dunlop
- T33) Dan Waschbusch
- T34) Michael Houlihan
- T35) David Shore
- T36) Sherry Stewart

3:00—3:30 **Coffee Break**

3:30—5:00 **Symposium "Psychology at Dalhousie: Where have we been and where are we going?"**
330 **Chair: Richard Brown**
- 355 Donald Mitchell: Perception and sensory neuroscience
- 410 Ray Klein: Cognitive psychology
- 425 Pat McGrath: Clinical psychology
- 440 John Connolly: Cognitive neuroscience
- 455 Chris Moore: Developmental psychology

Gala Celebration

6:30 pm	Cocktails
7:00 pm	Dinner at Faculty Club. Dance to follow.

All presentations are listed in the name of the first author from the abstracts submitted.

Welcome to the 1999 Psychology and Neuroscience In-House Convention

Welcome to the Twenty Fifth Annual Dalhousie University Department of Psychology In-House Convention. This year's meeting is being hosted by the students and faculty of our Graduate Program in Psychology. The program planning committee consists of the following individuals: N. Blanchet, R. Brown, W. Brown, V. Chu, C. Forestell, R. Newman, K. Shoemaker, and M. Yoon. A special thanks to Suzanne King for efforts above and beyond the call of duty, and David Rioux for his assistance with the In-House Convention web page and computer presentation.

Friendly Reminder

Please bring your own mug for coffee or tea!

An Historical Introduction to the In-House Conference

Richard Brown

This presentation will use photographs from the past 25 years to introduce the In-House. Photos of past faculty, students, and PDF's in their most embarrassing moments will be presented. To celebrate the 25th In-House, former students and faculty will provide comments on their memories of past In-House conferences.

T1. Pain, Pain Coping Strategies, Affective Distress, and Return to Function in Athletes Following ACL Surgery

Dean Tripp

This study has examined pain, coping, and functional recovery in athletes who have had Anterior Cruciate Ligament (ACL) reconstructive surgery. All patients were examined pre and post-op on a variety of measures. Post-op assessment periods were: 24, 48, and 72 hrs, 2, 8, 16 weeks. As may be expected pain and functional disability peaked shortly after surgery and gradually declined over the course of 16 weeks. The relationship between pain coping and effective distress however tells a much more interesting story... stay tuned.

T2. Electrophysiological Studies of Task Complexity: Does P3b Latency Reflect Stimulus Evaluation Time?

not given

Ivan Kiss, Wolf Fazikas, Hannah Pazderka-Robinson and Darlene Floden

P3B latency has been considered a measure of stimulus evaluation time (Kutas, McCarthy & Donchin, 1977; McCarthy & Donchin, 1980; Magliero et al., 1984). However, it is not clear which of the processes involved in stimulus evaluation are indexed by P3B; whether it reflects detection of salient stimulus features, completion of cognitive processes required for full target identification beyond the physical stimulus level or some intermediate stage.

We have completed two experiments in which task difficulty was manipulated with minimal changes to stimulus properties and response requirements. P3b latency measures and the emergence of additional ERP features confined to more difficult processing tasks suggest that:

1) P3b does not index completion of stimulus evaluation
2) Longer latency components emerge with more complex tasks.

We also report results of preliminary studies involving a "depth of processing" manipulation which suggest that P3b latency is sensitive to semantic processing requirements.

T3. Nitric Oxide and Anxiety in Rat Pups

Jana Podhorna and Richard E. Brown

The role of nitric oxide (NO) in anxiety was studied in the model of ultrasonic vocalization (USV) in rat pups using drugs that manipulate formation of NO in the brain.

Both the selective neuronal nitric oxide synthase (NOS) inhibitor (7-nitroindazole) and the non-selective NOS inhibitor (L-NAME) decreased USV in a dose-dependant manner. However, the non-selective inhibitor also suppressed locomotion. The inactive isomer of the NOS inhibitors, D-NAME, produced no effect on USV or locomotion. The NO precursor, L-arginine, had no effect on USV but increased locomotion at low doses.

Thus, drugs that decreased NO synthesis showed anxiolytic effect. However, only the selective NOS inhibitor had a selective anxiolytic effect, i.e., reduced USVs without causing sedation. Increased synthesis of nitric oxide in the brain had no apparent effect pro-anxiety effect but increased locomotion in this model of anxiety.

T4. An Investigation of Dissimulation in a Chronic Pain Sample

Lachlan A. McWilliams

Research aimed at detecting pain patients' dissimulated responses on psychological self-report measures has been conducted under the assumption that a small number of patients may intentionally exaggerate their psychological symptoms in order to receive increased financial compensation (e.g., continued disability benefits). In the present study, a sample (N=85) of chronic pain patients assessed at a multidisciplinary rehabilitation program was utilized to investigate the incidence of various self-presentation styles. Previous assumptions were not supported as scores on the validity scales of the Personality Assessment Inventory (Morey, 1991) were indicative of a high frequency of overly positive self-portrayals and the absence of unrealistically negative self-portrayals. Correlations between the validity scales and subscales of the Multidimensional Pain Inventory indicated significant negative associations between positive self-presentations and pain-related variables such as pain severity and pain-related distress. The implications of these findings for research and clinical practice will be discussed.

T5. On Earth as it is in Heaven: Trinitarian Influences on the Western Concept of Person

John Barresi

I will discuss a fragment of western intellectual history: how the concept of person developed from thinking about the relationship between God and human beings. In particular, Christian beliefs about God as a trinity of persons, and Christ as person, who is both God and man, had an enormous influence on the development of the concept of person in western thought. My interest, here, is in how our conception of our selves, both metaphysically and psychologically, interacted with our conception of God. Just as Jim Clark thinks that animals served, historically, as a mirror for human self-concepts, I think that God and the spirit world played a similar role. In this paper I will trace some of the historical developments that show this parallel genesis in conceptualization of human and divine personality.

T6. Using Mouse IQ to Assess Learning and Anxiety Behaviour in Barrelless Mice

Lianne Stanford (1), Richard E. Brown (1), Sean Taylor (2), and Paul E. Neumann (2), Departments of (1) Psychology & Neuroscience, (2) Pathology, & Anatomy & Neurobiology

The barrelless (brl) mouse strain was derived from a naturally occurring mutation observed in ICR mice at the Universite de Lausanne (Switzerland). brl/brl mice show a lack of specific cortical organization known as barrelfields. In control mice, barrelfields are formed by the whisker neuron thalamocortical organization. A high-resolution genetic map of the region around the brl locus revealed that the gene for adenylyl cyclase 1 (Adcy1) was missing in the mutant. Adenylyl cyclase 1 is associated with neuroplasticity in the hippocampus and cerebral cortex (Abdel-Majid et al. 1998. Nat Genet, 19, 289-91). Using targeted mutagenesis, and Adcy1 knockout mouse model was created that performed poorly in some measures of the Morris water maze task (Wu et al., 1995, PNAS, 92, 220-4). To further investigate the role of Adcy1 in memory processes, brl/brl and control (brl/- C57Bl/6J) mice of both sexes were studied using tests that measure locomotion and exploration (open field), anxiety (elevated plus maze), novel object recognition, and spatial learning (Morris Water maze). Differences between genotype and sex in a variety of measures on each test will be reported. This will provide insight into potential behavioural deficits due to the cortical misorganization present in the brl/brl mice. The importance of using multiple tasks to assess cognitive function in mice will also be discussed.

T7. Is Prospective Altruistic Detection an Evolved Solution to the Adaptive Problem of Subtle in Cooperative Ventures?

William M. Brown and Chris Moore

Reciprocal altruism in humans may be made possible in part by the existence of information-processing mechanisms for the detection of overt cheating. However, cheating may not always be readily detectable due to the division of labour. Subtle cheating poses a serious problem for the evolution of altruism. This paper argues that subtle cheating may have exerted selective pressures on early hominids to be sensitive to information regarding the genuineness of an altruistic act. In a two experiments, subjects were required to complete Wason selection tasks designed to allow for the detection of altruism. Performance on the altruist-detection tasks was compared to performance on control Wason selection tasks (Experiment 1) and to performance on control and cheater detection tasks (Experiment 2). Participants were significantly better at solving cheater detection and altruist detection versions compared to control versions of the problems and there was no significant difference between altruist and cheater detection. Results are discussed in relation to recent conceptual models for the evolution of altruism. Specifically, it is argued that non-kin altruism may be an evolutionarily stable strategy if altruists can detect one another and form mutually beneficial social support networks.

T8. Does Sex Make You Sick?: Trade-offs Between Immunity and Reproduction in the Cricket, *Gryllus integer*

Shelley Adamo

Animals have finite energy budgets. The immunocompetence hypothesis postulates that males will down-regulate their immune systems at the time of sexual maturity and divert this energy into reproduction. This theory is controversial, largely due to the difficulty in testing it in vertebrates. The theory can be better tested in crickets, which have simpler immune systems. I estimated the amount of energy invested in immunity during a male's lifespan by 1) counting the number of circulating immune cells and 2) by measuring the titres of an anti-microbial protein (prophenoloxidase). The ability of the immune system to destroy parasites and kill bacteria was also tested at different ages.

There was no evidence for a decline in energy invested in immunity in sexually mature males over their lifespan or relative to females. It is possible that immune system costs have already been minimized, and further reductions in immune capacity would lead to a shortened lifespan and reduced total lifetime fecundity.

T9. Role of Suppressed Feeding in *Manduca sexta* in Response to a Bacterial Infection.

Cathy Forestell and Shelley Adamo

Typically animals suppress their intake of food when sick. Traditionally, this was viewed as a maladaptive response since increased energy is required for an effective immune response. However, there is some evidence to suggest that infection-induced anorexia may aid the host in suppressing bacterial growth. It has been suggested that infected animals suppress their feeding to deprive invading bacteria of iron, an essential element for their survival. In this study, iron levels were manipulated in tobacco hornworms infected with the insect pathogen, Serratia marcescens. On day one of the fifth instar, animals were fed either an iron-rich diet, a diet with normal amounts of iron, or an iron-free diet and the effect on the amounts of free and bound iron in the hemolymph was assessed. Six hours later, these animals were injected with an LD50 dose of Serratia marcescens. It is hypothesized that the animals fed the iron-rich diet will show a greater mortality than those fed the normal iron diet, whereas those fed the iron-free diet will show lower mortality than those fed the normal iron diet.

T10. Adaptive Changes in Egg-laying in the Cricket, *Gryllus integer*, in Response to Infection by the Potentially Lethal Bacterium, *Serratia marcescens*

Kelly Shoemaker and Shelley Adamo

Animals must distribute resources to both somatic and reproductive activities, and the energy allocated to reproduction at each age is selected so that lifetime reproductive success is maximized. However, if the probability of future reproduction declines due to age or illness, then it would be adaptive for animals to re-allocate current resources, and increase reproductive effort at the expense of survival.

Previous tests of this hypothesis have been performed primarily on birds or mammals, where parental care complicates the issue. We are currently testing the hypothesis in the field cricket, *Gryllus integer*. *G. integer* females lay eggs, but do not exhibit parental care. Their life history suggests that they should increase egg-laying when infected with a potentially lethal pathogen, such as the bacterium, *Serratia marcescens*. Experiments are in progress.

T11. Timing of Transitions in Adolescence

Sonia H. Greaven, Darcy A. Santor, Vivek Kusumakar

The study of adolescent transitions, which mark the development of the individual from childhood to adulthood, must address the issue of temporality. Although the changes associated with adolescence have been shown to occur in an interrelated manner (e.g., Lerner et al, 1996), the effects of the timing of adolescent transitions has not been adequately explored. The prevalence of various transitions in a normal adolescent population (N = 596) was examined. The analyses of these data determine a) whether each of 12 transitions has been experienced by a majority of adolescents, and b) which transitions are viewed as most important by the adolescents. The effect of timing of transitions will be addressed in 3 ways. First, whether the timing of transitions in comparison with the adolescents' social referent will affect how the transitions are experienced. Secondly, whether particular subsets of these changes occur in an interrlated and/or sequential manner. Finally, the question of whether the lack of a given transition within a normative sequence represents a potential risk factor will be addressed.

T12. Issues of Confidentiality in Testimony Given by Psychotherapists

Nadine H. F. Rossy

The role of a psychotherapist is changing dramatically. Psychotherapists are professionals who examine issues that an individual may have through the use of a variety of techniques. In addition, a psychotherapist may act as a resource seeker, detective, confidante, mediator, and team coordinator. More recently, psychotherapists have acquired two roles in the legal arena. In the legal arena, the psychotherapist may act as an expert witness or a fact witness. As an expert witness the psychotherapist is asked to give his/her opinion about legal issues and/or may act as a forensic consultant. In contrast, as a fact witness, the psychotherapist testifies to the direct observation of the client during therapy. The issues of confidentiality, informed consent, and legal privilege that arise for a psychotherapist in the role of fact witness will be presented as well as alternative courses of action that may be taken. It is concluded that a need for education about this dilemma exists.

T13. The Role of Semantic Memory in Category-Specific Visual Agnosia

Karen E. Jaskie and Patricia A. McMullen

Category-specific visual agnosia (CSVA) is a disorder of object recognition in which the ability to recognize objects belonging to one or more categories is selectively impaired following brain damage. In one form of CSVA, the ability to recognize living things is impaired, while recognition of non-living things remains relatively intact. Case studies of CSVA have indicated damage to both semantic memory mechanisms and visual perceptual mechanisms. A variety of theories have been developed to explain the existence of a living/non-living double dissociation. McRae et al. (1997) emphasize the importance of intercorrelations amongst individual semantic features. Upon analyzing 190 different concepts, they found that there was a much higher intercorrelation between the features of living things than artefacts. Using a lexical decision priming task, we plan to investigate differences in the connectivity of the features of living things and the features of artefacts within the network of semantic memory.

T14. Orientation Effects in Face Recognition: How Similar Are They to Object Orientation Effects?

Mary-Ellen Large

A common debate in face and object recognition literature is whether faces and objects share similar processing systems or not. As an extension to this debate my proposed research investigates whether faces exhibit the same characteristic responses to orientation as objects. Joliceur (1985) found that the time to name rotated objects (between 0 deg and 120 deg) varies systematically as a function of departure from an upright orientation. With practice the magnitude of this orientation effect decreases significantly. Tarr and Pinker (1989) demonstrated, using novel abstract stimuli, that this effect diminishes because orientation-specific templates are formed each time the objects are identified. However, Murray et al (1993) failed to support this view when they used line drawings of real objects. Further work by Schellinck, McMullen and Hamm (manuscript) using feature scrambled objects suggests that practice effects are based on orientation-invariant features. The purpose of further study is to discover whether a practice effect occurs with photographic face stimuli. If a practice effect is found the next stage will be to investigate whether this phenomenon is based on orientation-specific templates or orientation invariant representations.

T15. Awareness and Implicit Memory Using Event Related Potentials During Recovery from Anesthesia in Adolescence

Deanna Braaksma, Allen Finley, Patrick McGrath, and John Connolly

Explicit and implicit memory has been shown after General Anaesthesia (GA) suggesting that information was processed deeply enough to access its semantic content. A late latency Event Related Potential (ERP) component, the N400, is sensitive to the semantic features of language and requires attention to both the eliciting stimulus and its context. In the present study, auditory ERPs to semantically congruent and incongruent word pairs will be used to assess unconscious cognitive processing and the presence of implicit memory during recovery from anesthesia. Adolescents who are booked for minor procedures under GA will be recruited from a Childrens Day Surgery ward. Eight channels will be recorded and word pair stimuli will be presented preoperatively and during and after recovery from GA. Implicit memory will be assessed using a word association task to primed words. It is hypothesized that the presence of an N400 will be related to the presence of implicit memory.

T16. Animate and Inanimate Entities

Nicole Blanchet

Traditional Piagetian theory has argued for a developmental change in children's early conceptual strategies called the thematic-taxonomic shift. In conventional match-to-sample and sorting tasks, young preschoolers have been described as employing thematic strategies (e.g. matching dogs with bones), while older preschoolers have been said to employ taxonomic strategies (e.g. matching dogs with dogs). More recent research has questioned this view, suggesting instead that children across this age range reveal a persisting tendency to organize their world into like kinds based on perceptual similarity. These more recent data have also provoked a new set of questions about the factors that might undermine the pervasive early influence of perceptual similarity, and facilitate the adoption of thematic conceptual strategies instead. In this presentation, I will describe preliminary plans for my thesis research in which I hope to examine some potentially differential effects of animate and inanimate stimuli on children's tendencies to adopt taxonomic and thematic conceptual strategies.

T17. Using Stimulus-response Compatibility to Stimulate Relative Contribution of Intention of Attention to Inhibition of Return

Jason Ivanoff & Raymond M. Klein

Two experiments examined the interaction between the Simon effect (SE) and inhibition of return (IOR). The SE refers to better performance when the task-irrelevant location of the imperative stimulus and response spatially correspond relative to when they do not correspond. IOR refers to better performance when the location of a precue and target do not correspond elative to when they do correspond, when cue location is task irrelevant and stimulus onset asynchrony (SOA) is generally greater than 400 ms. In the present investigation, the form of the precue provided subjects with response selection information, and the target provided response execution information. To the extent that IOR has a motoric component, inhibition at the cued location should reduce the SE. Similarly, this prediction is in line with an attention shifting account of the SE: there should be a smaller SE at the cued location relative to the cued location. At the long SOA, IOR was larger when the spatial location of the target and response corresponded relative to when they did not correspond. Similarly, SE was greater at the uncued than at the cued location. The magnitude of IOR was influenced by the intentions to produce a response. The results support an attentional-motor view of IOR and they support an attention-shift account of the SE.

T18. Cued Judgments of True and False Memory Reports: The Effect of Training and Individual Differences

Mary Ann Campbell & Stephen Porter

The false memory debate has highlighted the complexity of evaluating memory reports in both legal and therapeutic settings (e.g., "recovered" memories of abuse, erroneous eyewitness statements and false confessions). As such, the present study examined the influence of cue training and individual differences on the ability to make accurate credibility judgments of true and false memories. Information about the personality characteristics and baseline measures of perceived cues to false memory reports were obtained from the sample of 100 university undergraduate students. Half of the sample received information about cues research has shown to be characteristic of true or false memories (training group). A control group received no training. Both groups viewed a video of a common target recounting a truthful and implanted childhood memory in separate interviews and were asked to judge the credibility of both reports and indicate the cues used to assist their judgments. Judgment accuracy was assessed separately for true and false reports. Although results are not yet available, a significantly greater use of appropriate cues and better detection accuracy of the trained group compared to controls would support the hypothesis that training improved detection of false memory reports. The results of this study will offer insight into the benefits of training and the role of individual differences in judgments of memory credibility. Further, the findings will contribute to a framework on which the training of law enforcement professionals and therapists could be developed to assist in making more accurate credibility assessments.

inhibition attentional flexability

working memory

T19. Executive Function Abilities and Social Behaviour in Young Children

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Shana L. Nichols & Chris Moore

Interest in executive function abilities in young children has flourished within the area of developmental psychopathology. However, little is known about the normal development of executive functions and the relationship between executive function abilities in young chidren and social behaviour has been largely unstudied. Recently, Moore, Barresi, and Thompson (1998) found a relationship between children's performance on an inhibitory control task and their ability to engage in future-oriented sharing behaviour. Further research is needed in order to address the nature of executive function development in young children and its implications for emerging social behaviour. The goal of the present research was to explore possible links between executive functions and social behaviour in three-and-a-half year olds. Executive function tasks included measures of inhibitory control, attentional flexibility, and working memory. Social behaviour was examined in two ways: behavioural observations during a free play session and a future-oriented sharing task.

T20. Self-processes in Early-elementary Age Children

? **Sharon Clark** Cory Simmons

A sense of self is thought to emerge from early interactive relationships with significant others. Internal working models form two complementary models of self and other and include the child's beliefs concerning worthiness of care in the conceptualisation of self, and statements regarding the dependability and acceptance of the attachment figure in other. Concurrent and predictive relations of child-mother attachment security to self-concept and self-esteem in 29, 5-year-old children (girls n = 10, boys n = 19) were examined. Self-esteem was defined as the evaluative component of the self and self-concept as the descriptive component of self. Attachment security was measured at age 2 and at age 5. Children who were able to admit imperfections in themselves and maintained a positive valence on the self-esteem measure had higher attachment security scores at age five than at age two and had higher self-concept scores than the other children. Data currently being collected will be discussed along with these earlier findings.

T21. Event-Resolution Imaging: Overcoming the Limitations of Signal Averaging in ERP Research

? **Randy Lynn Newman**

The interpretation of event related potential data has typically relied on the process of signal averaging to extract meaningful information from background noise. The major limitation of this process is that numerous trials are required to generate a single meaningful waveform. A new method of interpretation developed by Thoughtform Corporation, called event-resolution imaging (ERI), appears to overcome this limitation. ERI employs a variety of methods to discover a set of features which best discriminates and identifies epoch types. This information is encoded as an interpretation map, which is subsequently used to analyze and interpret raw EEG data on an epoch by epoch (i.e. single trial) basis. The result of this analysis, is the production of a high signal-to-noise, non-averaged waveform. A demonstration of ERI will be performed, to be followed by a discussion of its potential role in current ERP research.

T22. Are There Differential Effects of a Selective Reminding Procedure on the Learning Characteristics of Individuals with Alzheimer's Disease and Isolated Memory Impairment?

Fernando A. Larrea and John D. Fisk

There is currently debate regarding the effect of cues on the learning of subjects with Alzheimer's Disease (AD). Some have suggested that cuing is particularly important in AD, whereas others don't support this facilitation effect. This study examined responses on the Buschke Cued Recall Test (BCRT) to compare free and cued recall in groups of subjects in the Canadian Study of Health and Aging with either AD or isolated memory impairment. Subjects with isolated memory impairment performed better than the AD subjects on all measures. However, both groups benefitted to a similar extent from cues on the initial learning trials and both had a similar rate of learning on the free recall trials. The quantitative but not qualitative differences in performance of the groups suggests that many subjects with isolated memory impairment are in the pre-clinical stages of AD. The positive learning slope for the AD group suggests that a cued recall procedure can facilitate learning in AD subjects.

Keynote Speech: The Discovery of Spoken Language

Dr. Peter Jusczyk

To acquire a native language, a learner has to correctly identify the units (words, phrases, and clauses) that are combined and manipulated in forming utterances in that language. A failure to identify the appropriate units in the speech input would greatly complicate learning how sentences are structured in the native language. There is considerable evidence that, during the second half of the first year, infants rapidly converge on the specific properties that distinguish the sound organization of their native language from other languages. Infants' developing sensitivity to native language sound structure not only plays a role in helping to segment words from fluent speech, but has consequences for learning about the language's grammatical organization. Thus from an early age, infants appear to use information in the speech signal to group the input in a way that facilitates the discovery of the structure of utterances. Findings from our laboratory show that infants detect cues to clausal and phrasal units in utterances well before they actually begin producing and combining words. Furthermore, the groupings that infants perceive in the input influence what information they encode and retain about speech. In addition, there is evidence that infants are sensitive to the sequential order of elements in the information that they retain. I will discuss some of these findings and their implications for understanding how learners discover the organization of their native language.

Jusczyk, P. W. (1997). The discovery of spoken language. Cambridge, MA: MIT Press.

Cognitive Clinical Neuroscience Unit

T23. An Investigation of Sentence Reading Using Event-related Functional Magnetic Resonance Imaging (fMRI)

Ryan C.N. D'Arcy, John F. Connolly, Larry Gates, and Elisabet Service

Neurophysiological studies of reading have successfully elucidated electrical and magnetic correlates of cognitive processing, such as semantic analysis manifested by the N400. We used an established paradigm from ERP/MEG research to study the neuranatomical regions of activation during sentence reading. The participants read a series of sentences with terminal words that were either semantically congruent or incongruent (e.g., The pizza was too hot to: eat/sing). In both conditions, activation was observed in those neuroanatomical regions (e.g., visual cortex) thought to be involved in reading. However, additional activation was observed in several regions, including the left angular gyrus, for the semantically incongruent condition. The findings concurred with previously localized N400 dipoles using MEG with the same experimental paradigm.

T24. Looking at Looking by Mind and Brain

Ray Klein and the SOPOS collaboration

Cognitive neuroscience is concerned with how the mental representations and processes that underly complex behavior are implemented by neural structures and processes in the brain. Our interdisciplinary team focussed on the control of gaze direction (overt visual orienting), in particular the control of fixation and saccadic eye movements, as a "model system" for exploring the interplay between volitional and reflexive control systems, the mechanisms of preparation, the influence of objects on actions. With a few examples, I will illustrate our interdisciplinary approach, which is characterized by a joint consideration of normal performance, the behavior of neurons during normal performance and the changes in performance that occur as a result of manipulations of (e.g. damage to) brain structures. Our understanding of how the brain controls looking is both reflected in and driven by our neural field model which is able to simulate oculomotor performance and the behavior of neurons in a wide range of tasks. An asymmetry in children's comprehension of verbal ironic criticisms and compliments.

T25. An Asymmetry in Children's Comprehension of Verbal Ironic Criticisms and Compliments

Jeff Hancock

Extant research on children's comprehension of verbal irony has focused on critical forms of irony to the exclusion of less typical complimentary forms. Two experiments examined children's ability to detect and comprehend ironic criticisms and compliments presented in videotaped stories. In experiment 1, children more frequently detected the non-literal nature of ironic criticisms than ironic compliments (first-order attribution). However, children that successfully detected the non-literal nature of an ironic statement were able to comprehend the intended meaning of both types of irony equally well (second-order attribution). Experiment 2 employed the same procedure but also manipulated the degree to which ironic statements echoed the previous statement in the story. A similar pattern of results was obtained, although this manipulation reduced the asymmetry in detecting the non-literal nature between ironic criticisms and compliments. The asymmetry across these two exemplars of irony suggests that a single underlying cognitive development (e.g., second-order attribution) may be a necessary condition for the detection of irony, but not sufficient.

T26. Stripping Away a Layer of Perception: Binaural & Monaural Spatial Channels

Susan E. Boehnke & D.P. Phillips

Two of the major perceptual dimensions of a sound are its frequency content and its spatial location. While the structure of the perceptual channels mediating frequency processing are well understood, the perceptual channels used for sound localization are not. We have used gap-detection methodologies to investigate this. The size of gap-detection thresholds can indicate whether sounds on either side of the silent period (gap) are activating the same or different perceptual channels. We varied speaker locations of the leading and trailing gap markers around the head, and obtained gap thresholds. The results supported a model of two broadly-tuned hemifield channels overlapping through the midline. The next step is to examine the channel tuning under monaural conditions in the same listeners (repeating the experiment with one ear plugged). This eliminates the binaural cues that contribute to sound localization, viz., the interaural differences in the arrival time and intensity of a sound. That is, ear plugging removes a layer of processing contributing to our perception. Results from this monaural study will be presented.

T27. A Controlled Trial of Interpersonal Therapy and Sertraline in Adolescents with Depression

Darcy A. Santor and Vivek Kusumakar, Department of Psychology and Department of Psychiatry, Dalhousie University

Interpersonal Therapy was initially developed as a brief, focused treatment to alleviate symptoms of depression and address current interpersonal problems experienced by individuals with depression. Although several studies have shown that IPT is effective in treating depression in adults, only one open trial of IPT for the treatment of unipolar depression in adolescents has been conducted. However, there are some concerns with this study, namely that treatment was provided by just one highly-trained clinician and that the sample in which effects were observed was extremely homogeneous. To address some of the concerns with the original open trial conducted, we conducted an open trial of IPT with a sample of 24 depressed adolescents who were matched on age and gender to a sample of 24 depressed adolescents receiving Sertraline. IPT was provided by 10 therapists, trained specifically in Interpersonal Therapy, each of whom saw between 1 and 3 clients. Results showed young persons improved substantially whether they received IPT or Sertraline; however, young persons receiving IPT improved more.

focus on · disputes · transitions · grief · deficits
Sertraline.
[illegible]
Rate. Stress
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T28. Nicotine, Muscarine and Circadian Rhythms

Shui-Wang Ying and Benjamin Rusak

There has been a long history of controvery about how the neurotransmitter acetylcholine affects circadian rhythms, and especially what role it plays in the synchronization (entrainment) of these rhythms by environmental lighting cycles. Some studies reported that cholinergic drugs act via nicotinic receptors to mimic the effects of light while others reported that nicotinic receptors were not relevant, but that muscarinic receptors mediated cholinergic effects on rhythms. In a combination of in vivo and in vitro neuro-physiological studies of cells in the suprachiasmatic nucleus, we have examined how each of these classes of cholinergic receptors interact with excitatory and inhibitory amino acid transmitters involved in circadian rhythm entrainment. Actions on nicotinic receptors appeared to be presynaptic and involved in the regulation of release of glutamate and GABA reaching SCN cells from other neurons, while actions via muscarinic receptors appeared to be postsynaptic; i.e., directly on SCN neurons.

T29. Molecules of Synaptic Transmission

Alfredo Villarroel, Department of Physiology and Biophysics

Synaptic transmission is the process used by neurones to communicate with other neurones and skeletal muscle. During synaptic transmission, a neurotransmitter released at a nerve ending diffuse to the synaptic space to reach receptors at the target cell. There it binds to ligand-gated channels, the neurotransmitter receptors that mediate fast-synaptic transmission.. During the last decade, research has revealed the process of ion movement through these channels. Initially, computer models, which showed the interactions between ions and the channel protein, guided the design of mutation. The study of mutated channels elucidated the process of ion selectivity, the process by which a channel can distinguish among ions. Pore size well as polarity of the filter determined ion conduction. Research on acetylcholine, glutamate (NMDA as well as AMPA/ Kainate) receptors, however, has not yet uncovered the mysteries of Ca conduction. Presently, I am interested in the study of natural mutation that occur in some neurological diseases.

T30. Why You Get Sleepy When You Stay Awake Too Long: The Role of Adenosine

Leticia Materi, Doug Rasmusson and Kazue Semba, Depts. of Anatomy & Neurobiology, and Physiology & Biophysics

The release of the neurotransmitter acetylcholine (ACh) is associated with an increase in the excitability of cortical neurons, as during wakefulness in general, and in attentive states in particular. However, as we stay awake and vigilant for a long period of time, we get drowsy. What causes this? To obtain clues to answer this question, we have investigated how the neuromodulator adenosine, which is released by brain cells during increased metabolism, affects ACh release, using in vivo microdialysis in anesthetized rats. We found that adenosine inhibits ACh release evoked by stimulation of brainstem reticular formation. This inhibition was concentration-dependent, mimicked by adenosine A1, but not A2a, receptor agonist, and blocked by caffeine, a non-selective adenosine receptor antagonist. These results suggest that, during prolonged wakefulness, adenosine released as a result of increased neuronal activity could diminish ACh release, inducing global inhibition of the cerebral cortex and a decrease in vigilance.

T31. In One Ear and Out the Other? Sound Transmission in a Generalized Bug Leg

Steve Shaw

Insects such as crickets bear specialized ears complete with ear drums (tympana) on their front legs, and use them to locate distant singing mates. It is therefore behaviorally advantageous for these ears to be strongly directional, and the directional sensitivity of the cricket ear at around 5 kHz is better than 20 dB (>10:1 max-to-min sensitivity ratio). Insects are rather small to achieve directionality by using sound-shadowing or time delays, and crickets do it by an interference mechanism, reinvented more recently by humans for directional microphones. Sound traveling outside the body to the front door of the tympanum interferes with (physically cancels) sound traveling down enlarged tracheal air tubes inside the body to the back door of the tympanum. Internal transmission is effective even via a large trachea crossing from the opposite side of the body. Directionality is conferred because cancellation varies in effectiveness with position, as the bug rotates itself relative to the sound source.

Specialized mechanisms usually don't just emerge *de novo*, but evolve from some related but more general precursor state present in an ancestor -- a preadaptation. Vibration detectors are present in the tibia of each leg below the knee, forming the subgenual organ (SGO), from which, in crickets in early development, the sensory neurons bud off that will later form the ear. Cockroaches, an ancient ancestor-model group, have been examined physiologically to see if some parts of the directional mechanism are recognizable in a more general insect leg, since the roach SGO also responds to sound, best around 2 kHz. The results presented from semi-intact roaches will show that sound can indeed travel down the leg's tracheae to the SGO, that this can interfere with and completely cancel sound entering through the leg's surface, and that low frequencies can even spread across the midline from the opposite leg, within the tracheal system. Much of the groundwork for response directionality must have been laid down quite early on in insect legs, which crickets later took over and refined.

T32. Left Limb Movement Improves Visual Scanning in Hemispatial Neglect

Beverly ~~Dunlop~~ Butler(1,2), Gail Eskes(1,2), Ed Harrison(1,2), & Alison MacDonald(2), Dalhousie University(1) and QE II Health Sciences Centre(2), Halifax, Nova Scotia.

Hemispatial neglect is a deficit in directing attention/movement toward the contralesional space. Robertson and North (1992;1993) reported in case studies that active left-limb movement in left hemispace can improve visual scanning in neglect, but that passive left-limb movement was not effective. We attempted to confirm and extend the findings regarding passive limb movement using functional electrical stimulation (FES) to produce passive movement. Subjects with neglect (n=6) detected visual targets under conditions of no movement and passive left-limb movement in left hemispace. Three non-paretic subjects also participated in an active movement condition. The passive movement group showed an improvement in target detection on the left ($p<.05$), while 3/3 subjects improved with active movement. Intentional motor programming does not appear required for activating perception-action circuits involved in attention. In addition, FES may have a role in the rehabilitation of hemispatial neglect.

T33. The Criterion Validity of Parent Ratings of their Child's Cognitive Abilities

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Examined the criterion validity of parent ratings of their child's cognitive abilities. Participants were 192 children referred to an outpatient mental health clinic for cognitive testing. Criterion measures were children's scores on the Woodcock-Johnson Psycho-Educational Battery - Revised and the Wechsler Intelligence Scale for Children - Third Edition. Results supported the validity of parent ratings of their child's fluid reasoning, visual-motor processing and verbal-auditory processing. Parent ratings of comprehension-knowledge and memory did not perform well. The findings provide evidence that parents may be accurate reporters of some of their child's specific cognitive abilities.

T34. Nicotine: How Does it Affect Performance in a Memory Scanning Task?

Michael E. Houlihan

Nicotine absorbed from cigarette smoke improves performance on many cognitive tasks. Although overall response times generally are faster after smoking/nicotine, few studies have tried to isolate the effect of smoking/nicotine on the speech of the short-term memory (STM) scanning substage of information processing. The present study investigates the effect of smoking/nicotine on STM scanning using both reaction time (RT) and event-related potentials (ERPs). An STM scanning task was performed before and after smoking each of two cigarettes. One cigarette had a 0.05-mg nicotine yield and the other had a 1.1-mg yield (FTC method). Two, three or four consonants were displayed as a memory set. After a brief interval, a single probe consonant was displayed. If the probe was in the memory set (positive probe), a right button press was required and if the probe was not in the memory set (negative probe), the left button was pressed. RT was faster after smoking the 1.1-mg cigarette but not after smoking the 0.05-mg cigarette. However, memory-scanning speed, as estimated from the increase in RT as a function of increasing set size, was not affected by nicotine. Smoking the higher-yield cigarette reduced N1 latency to the negative probes (at Cz) and also reduced N2 latency to both the memory set and negative probes. These ERP latency effects were small compared to the effects of nicotine on RT suggesting that nicotine shortens RT by affecting response-related processes. Nicotine produced a sustained positive shift in ERP amplitude beginning around 100 ms post stimulus.

T35. Of Mice and Men: Species & Individual (Human) Differences in Wayfinding Using a Virtual Hebb-Williams Maze

David I. Shore (1), Lianne Stanford (1,2), Joseph W. MacInnes (3), Raymond M. Klein (1,2) & Richard E. Brown (1,2). (1) Psychology Department, (2) Neuroscience Institute, (3) Daltech, Faculty of Computer Science

The Hebb-Williams maze is a standardized set of mazes developed to test animal spatial "intelligence". It has been used for over 50 years to compare the spatial intelligence and learning ability of many species including rats, cats, dogs, cows, ferrets, and fish. Despite being used a wide range of species, the Hebb-Williams maze has never tested spatial learning in humans. This may be because of the difficulty in designing a large enough testing apparatus, however, researchers have tested cows & sheep! Our project attempts to assess human performance using a computer generated three-dimensional virtual environment. As in many contemporary video games, participants take the perspective of navigating within a human-sized maze and are required to complete the standardized training and testing protocol typically administered to other species. The primary measures are reaction time to complete each of the twelve test mazes and the number of errors committed. Three independent measures of spatial ability were also collected: the Santa Barbara Sense-of-Direction Questionnaire (v.2), the WAIS-R block design test, and performance on a single Porteus maze. The results of performance on the maze are examined with respect to scores on these other tests of spatial ability and will be compared with results from other species. There are many areas for future applications and research using our Virtual Maze, including topics as broad as cognitive development, neuropsychology and rehabilitation assessment, individual and gender differences, and comparative psychology.

T36. Relations Between Personality Domains and Drinking Motives in Young Adults

Sherry Stewart & Heather Devine

The purpose of the present study was to place drinking motives within the context of the Five-Factor Model of personality. Specifically, we sought to determine whether certain personality domains of the Revised NEO Personality Inventory (NEO-PI-R; Costa & McCrae, 1992) predict Enhancement, Coping, Social, and/or Conformity drinking motives from the Revised Drinking Motives Questionnaire (DMQ-R; Cooper, 1994). A sample of 256 university student drinkers (M age = 21.3 years) completed the NEO-PI-R and DMQ-R. In bivariate correlations, the two negative reinforcement motives (Coping and Conformity) were positively correlated with Neuroticism and negatively correlated with Extraversion, whereas the two positive reinforcement motives (Enhancement and Social) were positively correlated with Extraversion and negatively correlated with Conscientiousness. Multiple regression analyses revealed that personality domain scores predicted two of the four drinking motives (i.e., the internal drinking motives of Coping and Enhancement), after controlling for the influences of alternative drinking motives. Enhancement Motives were predicted by high Extraversion and low Conscientiousness, and Coping Motives by high Neuroticism. These results provide further validation of Cox and Klinger's (1988; 1990) 2 x 2 (valence [positive vs. negative reinforcement] x source [internal vs. external]) model of drinking motivations, and confirm previous speculations that drinking motives are distinguishable on the basis of higher-order personality domains. Understanding the relations between personality and drinking motives may prove useful in identifying young drinkers whose drinking motivations may portend the development of heavy and/or problem drinking.

Posters

P1. Profiles of Psychopathy in Incarcerated Sexual Offenders

Steve Porter, Angie Birt, & Mary-Anne Campbell

We investigated whether psychopathy would contribute to an understanding of the nature and heterogeneity of sexual violence. Using the Psychopathy Checklist-Revised (PCL-R; Hare, 1991), level of psychopathy, callousness (Factor 1), and antisocial patterns (Factor 2) were examined in a diverse sample (N = 329) of incarcerated adult male sex and non-sex offenders. The offenders were categorized according to their crimes (*extra-familial molesters, intra-familial (incest) molesters, mixed offense molesters, rapists*, *mixed rapist/ molesters* and *non-sex offenders*). Mixed rapist/ molesters, rapists, and non-sex offenders scored significantly higher on the PCL-R than molesters, although all sex offender groups showed elevated Factor 1 scores. Most (64%) mixed rapist/ molesters were psychopathic, indicative of a "sexual psychopath" whose thrill-seeking is focussed on diverse sexual victims. There were different interpersonal and social deviance profiles among the groups and an interaction between factor scores and offense type. These profiles have implications for recidivism, treatment, and, more generally, a comprehensive theory of sexual violence.

P2. Profound Amnesia Does Not Impair Performance on 36-item Digit Memory Test – A Test of Malingered Memory

Ryan C.N. D'Arcy and Jeannette McGlone

Global amnesia spares immediate span of attention while compromising retrieval after a delay filled with distractions. Accordingly, we hypothesized that the Digit Memory Test (DMT) would classify persons with severe, disabling amnesia as "non-malingerers" better than malingering tests utilizing interference or recall paradigms. Fourteen cases with profound amnesia, including 2 persons with global amnesia, obtained scores of 100% correct on the 36-item DMT. However, the Rey 15-item Test of malingering and a multiple choice delayed recognition memory test yielded false positive outcomes. We conclude that the DMT has excellent specificity with respect to correctly identifying persons with severe memory disorders due to neurological lesions by correctly classifying all as "non-malingerers".

P3. The Spatial Localization Deficit in Visually Deprived Kittens: Comparison with Human Amblyopes

G. Gingras, D.E. Mitchell, R.F. Hess; Dalhousie University, Halifax, Canada; McGill University, Montreal, Canada.

Purpose In order to provide data that could assist resolution of a debate over the nature of the underlying neural deficit in amblyopia, we measured the deficits in spatial localization of visually deprived kittens by use of the same tests as those employed for the assessment of human amblyopes. Methods. The tests were conducted on one normal and 6 selectively deprived kittens. Four of the kittens were monocularly deprived by eyelid suture while on two others esotropia was induced by section of the lateral rectus muscle of one eye when either 17 or 29 days old. Of the 4 monocularly deprived kittens, 2 were deprived from near birth to 60 (C773) and 82 (C772) days of age while the others were deprived for either 6 (C791) or 30 (C775) days beginning at respectively, 35 and 38 days. When the animals were 5 months of age, measurements were made of the accuracy with which they could detect a misalignment between three Guassian blobs of high contrast. The outer two blobs were in vertical alignment and measurements were made, by use of a jumping stand and a method of descending limits, of the minimum detectable horizontal misalignment of the middle blob for both the deprived (or deviating) and nondeprived eyes. Measurements of alignment accuracy were made at different spatial scales (where the stimuli were spatial scaled versions of one another) and as a function of contrast. Results. For both the normal animal and the nondeprived eyes of each of the visual deprived kittens, there was a comparable proportional relationship between alignment accuracy and Guassian blob size to that observed in humans. Moreover, there was a similar weak dependency on stimulus contrast. As with amblyopic humans, the deficits in alignment accuracy were scaled in proportion to blob size but were in general larger than those reported in human amblyopes. The deficits were greatest in the strabismic cats where the alignment accuracy of the deviating eye was 30 times worse than that of the other eye. The deficits in the deprived eye of the monocularly deprived animals were not related in any simple way with either the length of the deprivation or the deficits in grating acuity. Conclusions. The similarity of the pattern of results to that observed in human amblyopes suggests that by suitable titration of the timing of deprivation it will be possible to produce deficits similar to those exhibited in human deprivation and strabismic amblyopia thereby allowing for converging information on the nature of the underlying neural deficits.

P4. Inhibitory Control in Children With Attention Deficit Hyperactivity Disorder

E.N. McLaughlin1, R.M. Klein1, & D.P. Munoz2. 1Dept. Psychology, Dalhousie University, Halifax, NS. 2Dept. Physiology, Queens University, Kingston, ON.

In an effort to fully explore the validity and nature of the recently proposed inhibitory control deficit theory of attention deficit hyperactivity disorder (ADHD; cf Barkley, 1998), we tested children with and without ADHD on five different tests of inhibitory control. The Colour-Word Stroop test assessed negative priming and inhibition of distracting information. A Simon/Erikson task tested filtering ability and inhibition at the response selection stage. A delayed eye movement task examined the ability to suppress reflexive and memory-guided saccades. Inhibition of return was measured in a covert visual orienting task. A stop-signal task measured the ability to inhibit a visually-triggered response. We will report on the degree to which children with ADHD differ from controls on the measures of inhibition derived from these tasks, as well as on the degree to which these inhibitory mechanisms tend to covary.

P5. Where do Fears of Publicly Observable Symptoms of Anxiety Belong: Anxiety Sensitivity or Fear of Negative Evaluation?

Lachlan A. McWilliams, Paula S.R. MacPherson, and Sherry H. Stewart

Current conceptualizations of anxiety sensitivity (AS) posit that AS is comprised of several lower-order factors including fears of physical, cognitive, and publicly observable symptoms of anxiety. Attempts to refine the AS construct have led to speculation that the fear of publicly observable symptoms component of AS may be more appropriately conceptualized as belonging to the domain of Fear of Negative Evaluation (FNE). The test this hypothesis, 216 university students were administered the Anxiety Sensitivity Index and the Brief Fear of Negative Evaluation Scale and a principal-components analysis with oblique rotation was performed on their responses to this pool of items. Four factors representing FNE and three AS components (viz., fears of physical, cognitive, and publicly observable symptoms of anxiety) were extracted indicating that fears of publicly observable symptoms of anxiety are distinct from FNE. A second principal-components analysis indicated that these four factors comprise a single higher-order factor.

P6. The Relationship Between Anxiety Sensitivity Scores of Parents and Their Adult Children

Susan E. Buffett-Jerrott, Sherry H. Stewart, Margo Watt and Kerry L. Jang

Calamari et al. found a strong positive relationship between the anxiety sensitivity scores of parents on the ASI, and their young children on the CASI. In a twin study, Stein et al. found evidence for heritability on the social and physical subscales of the ASI but not on the psychological subscale. The present study examined ASI scores of 160 university students and their parents. Results indicate that total ASI scores of parents and their adult children are marginally related. Consistent with the findings of Stein, there are significant positive relationships for the physical and social subscales, but not the psychological subscale. Overall, the results are consistent with Calamari's findings that ASI scores of parents and their children are correlated. However, the magnitude of the relationship is much smaller in the present study. This is likely due to the many intervening events besides parental influences that may affect the development of anxiety sensitivity in individuals as they get older.

P7. Neuropsychological Predictors of Quality of Life in Stroke Survivors

Anne Israeli and Gail Eskes

The current study was conducted to assess whether cognitive function and activities of daily living, as measured using the Neurobehavioral Cognitive Status Exam (NCSE) and Barthel Index (BI), would predict short-term outcome following stroke. Forty-two patients with first stroke completed the NCSE within two weeks of admission to an acute stroke service. Discharge BI scores were calculated 2-4 days prior to discharge. Three months post-stroke outcome was measured with the BI, the Reintegration to Normal Living Scale (RNL) and the Medical Outcomes SF-36 (MOS-SF-36), obtained via telephone interview. A series of stepwise regressions revealed that discharge BI scores and the Orientation subscale of the NCSE predicted follow-up BI and RNL scores, but did not predict MOS-SF-36 scores. During the acute phase following stroke, basic cognitive processes such as orientation and activities of daily living may provide relevant information regarding short-term outcome.

P8. Increased Processing Demands During Target Error Detection on the Computerized Token Test

Ryan C.N. D'Arcy and John F. Connolly

Recently, we have adapted the Token Test (De Renzi and Vignolo, 1962, Brain, 665-678) for computer presentation with simultaneous event-related brain potential (ERP) recordings. The Token Test is a standardized neuropsychological measure of auditory comprehension. The objective of this study was to investigate whether variations in stimulus complexity influenced the cognitive processes involved in spoken word identification. The computer Token Test (CTT) was designed as a target detection task that required individuals to understand spoken sentences in order to detect an incorrect target word. For each trial on the CTT, participants were instructed to study a short animation sequence presented on a computer monitor. The animation sequence consisted of a pointer that touched either one or two tokens. Immediately after the animation, participants heard a single sentence description that was either correct or incorrect. ERPs were recorded to the onset of incorrect target words within the spoken sentences. In the Similar condition, the initial phonemes of the incorrect words were similar to the initial phonemes of the correct words (e.g., circle/square). In the Different condition, the incorrect target words had different initial phonemes and thus could be identified as incorrect at word onset (e.g., green/red). The ERP responses to words in the Similar condition elicited a negativity in the 250ms to 350ms region, which was not observable in the ERP responses to incorrect words in the Different condition. The negativity was hypothesized to reflect increased processing demands during error detection when the initial phonemes of the target words were similar.

P9. Left Temporal Cortex Activation Reflects Distinct Time Courses of Word and Context Comprehension

J.F.Connolly, P.Helenius, E.Service, and R.Salmelin

The time course and cortical basis of reading comprehension were tracked using magnetoencephalography (MEG). The cortical structures implicated most consistently with comprehension were located in the immediate vicinity of the left auditory cortex, where final words totally inappropriate to the overall sentence context evoked enduring activation between 250 ms and 600 ms after word onset. Contextually appropriate but unexpected words produced weaker activation, which terminated earlier. We propose that the point in time (350 ms after word onset) where the response to appropriate but unexpected endings started to diverge from those to contextually inappropriate endings reflects the boundary between understanding a single word and the meaning of a whole sentence.

P10. Neuropsychological Assessment of Receptive Vocabulary and Comprehension With Event-related Brain Potentials

John F. Connolly, Ryan C.N. D'Arcy, Alma M. Major, Sheri L. Allen, Gail A. Eskes, & Stephen J. Phillips

The clinical application of evoked potentials has proven invaluable for assessing sensory functions. Recently, we have adapted standardized neuropsychological tests for computer presentation with simultaneous event-related brain potential (ERP) recordings for assessing cognitive functions. The Vocabulary and Similarities sub-sections of the Wechsler Adult Intelligence Scale-Revised as a Neurological Instrument (WAIS-R-NI) were used to assess receptive single word vocabulary and verbal concept formation, respectively. The Written Comprehension section of the Psycholinguistic Assessments of Language Processing in Aphasia (PALPA) was used to assess receptive sentence comprehension. Test items from all three paradigms consisted of a study phase and a test phase. During the study phase, the test question and response alternatives were presented on a computer screen. Participants were requested to study the question, select an answer, and remember the answer. During the test phase, the response alternatives were presented individually and ERPs were recorded. Participants were requested to identify the correct and incorrect response alternatives. Items which the individual indicated as 'correct' were differentiated from alternative items by the presence of a P300 response. The P300 response was hypothesized to represent recognition memory and contextual updating processes following the presentation of a meaningful stimulus. Receptive language dysfunction is frequently noted in aphasic individuals suffering from stroke or traumatic brain injury. However, due to communicative limitations, these patients cannot be readily assessed with traditional methods. In these and similar instances, it is possible to assess language and other cognitive functions with neural responses that are relatively independent of behaviour.

P11. Differences in Children's Conceptual Strategies When Thinking About A Century of Rats in Mazes

Robert L.H. Clements and Richard E. Brown

Between 1898 and 1900, Willard S. Small studied development and general behaviour in rats at Clark University. In order to quantify their learning ability, he tested them in a rat-sized model of the Hampton Court Maze. With this study, Small (1901, The American Journal of Psychology, 12:207-239) was the first to combine laboratory investigation using rats and the concept of an intellectually puzzling maze. Over the last one hundred years, a wide variety of mazes have been used to study basic questions regarding the sensory and neural basis of learning, motivation and perception.

A multitude of species have been tested in mazes, from worms to humans. Innumerable mazes have been developed and used widely in psychological research, including the T-maze, Lashley's type 3 maze, Tolman's sunburst maze, the Hebb-Williams maze, Barnes Maze and the 8 arm radial maze. Mazes such as the Porteus maze, developed in 1917, and newer computerized virtual mazes, are still used as standardized tests for human mental abilities.

The current literature lacks a comprehensive overview of the use of mazes in scientific research. Our paper provides a history of this fascinating apparatus and its use. An in-depth review of the literature found over 11,000 papers on maze learning in the last 100 years and we are going to tell you about every last one of them, with particular emphasis on some of the currently popular mazes used in psychology (Morris water maze, 8-arm radial maze, elevated plus maze). The discussion summarizes the advantages and disadvantages of using mazes in psychological and medical research.

P12. Olfactory Conditioning and Differential Immediate Early Gene Expression in Mice

Catherine A. Forestell,1 Heather M. Schellinck,1 Richard E. Brown,1 Vincent M. LoLordo,1 Michael Wilkinson2 1 Department of Psychology and Neuroscience 2 Obstetrics and Gynaecology, IWK Grace Health Centre

The present experiment investigated whether adult mice acquired conditioned responses to an odour paired with a caloric reinforcer and which parts of the brain are involved in the formation of this association by looking for differences in c-fos-like-immunoreactivity. Food restricted mice were exposed to a differential conditioning procedure in which experimental animals received an odour (CS+) paired with sucrose pellets in a dish covered with bedding and another odour (CS-) presented alone in a dish with bedding. Control animals were either exposed to the odours separately in a dish containing bedding and no reinforcement, or to no odour. After 32 conditioning trials, animals were placed in a chamber divided into three equal sections with CS+ and CS- on each end of the chamber. Experimental animals spent significantly more time digging in the dish containing CS+ than CS- relative to control animals. On the next day, all animals received exposure to CS+, CS-, or no odour for 10 min, 2 hours prior to perfusion. Fos-like-immunoreactivity was assessed in brain structures such as the glomerular layer of the olfactory bulbs, piriform cortex, the hippocampus, and the periventricular nucleus of the thalamus.

P13. Effects of Methylphenidate Hydrochloride (Ritalin) Injections During Development on Locomotion, Exploration, Fear, and Cognitive Processes in Mice

Melanie McFadyen, Norman Carrey, and Richard Brown

Although the use of the stimulant methylphenidate (Ritalin) is increasing among children, little is known about the potentially deleterious effects it may have on their cognitive and behavioural development. Our study uses a mouse model to examine the developmental effects of methylphenidate (MPH) (daily 40 mg/kg sc vs. saline injections) in male CD-1 mice, from postnatal days 26-32. The mice were tested from postnatal days 33-37 for locomotion and exploration in the open field (day 1), anxiety in the elevated plus maze (day 1), and learning the Morris water maze (days 2-5), with a visible platform for three days and a hidden platform on day 4. Our results indicate that MPH-treated mice were more active in the open field, entering more centre squares than the saline controls; MPH-treated mice spent more time in the open arms and exhibited more head dips in the elevated plus maze than controls, and that, while there was no significant difference between MPH and saline-treated mice in the time taken to find the visible platform in the Morris water maze, the MPH-treated mice showed a tendency to take longer to find the hidden platform. The results indicate that treatment with methylphenidate has significant effects on later anxiety, reducing fear and increasing locomotion, but less of an effect on learning in a swimming task. However, since the methylphenidate-treated mice were hyperactive, their learning may be affected more in tests of inhibition.

P14. A Factor Analytic Investigation of the Alcohol Expectancy Questionnaire -3rd Version

Alan B. MacDonald and Sherry S. Stewart

The Alcohol Expectancy Questionnaire - 3rd version (AEQ-III; Goldman, Greenbaum & Darkes, 1997) is a self-report instrument consisting of 68 items designed to assess positive expectancies or beliefs about the outcomes of drinking alcohol. The AEQ-III is a revised version of the original, 90-item AEQ, and has not yet been evaluated extensively from a psychometric perspective. The purpose of the present study was to explore the factor structure of the AEQ-III in an undergraduate university sample and attempt to replicate the factor structure found by Goldman et al. (1997). Eight-hundred and nine undergraduates students were administered the AEQ-III. Exploratory principal components analysis with oblique (Oblimin) rotation identified that six factors best explain the underlying dimensions of this measure in this population. These factors include: 1) Social Assertiveness (SA); 2) Global Positive Changes (GP); 3) Social & Physical Pleasure (SP); 4) Sexual Enhancement (SE); 5) Arousal/Aggression (AA); and 6) Relaxation (R). The factor structure proposed by Goldman et al. (1997) was replicated. Implications for the use of this measure in determining potential alcohol expectancies and patterns of drinking in young adults are discussed.